

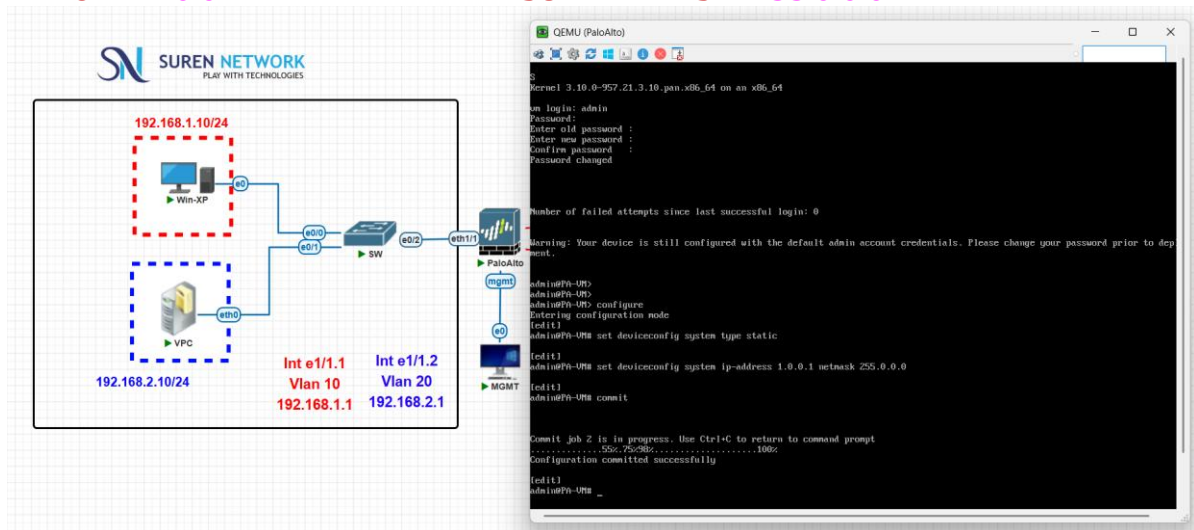
LAB 2 ECMP INTERVLAN

GURUCHANDHRAN M

TASK 1: FIREWALL IP ASSIGNMENT

PALO IP: 1.0.0.1

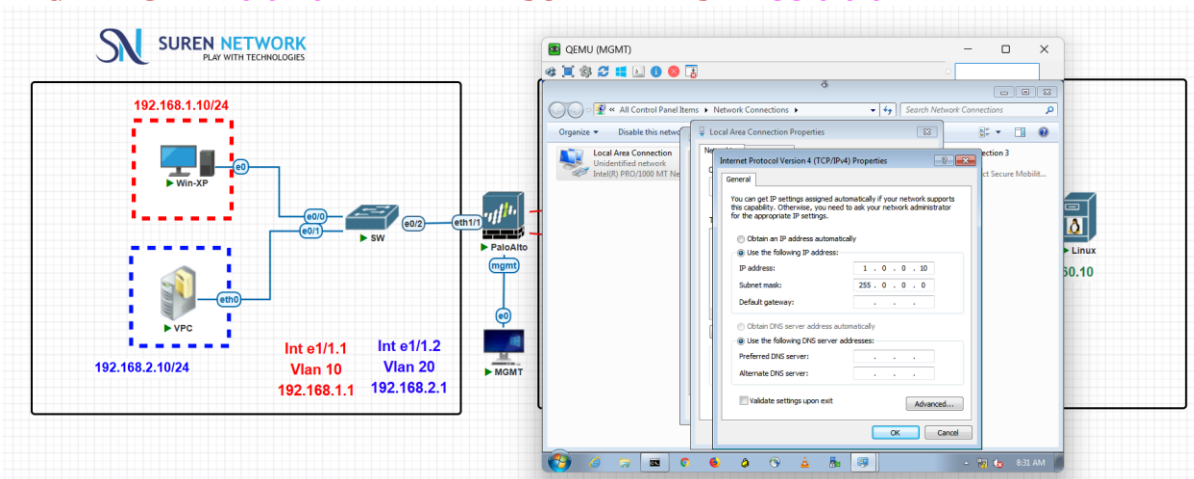
SUBNET MASK: 255.0.0.0



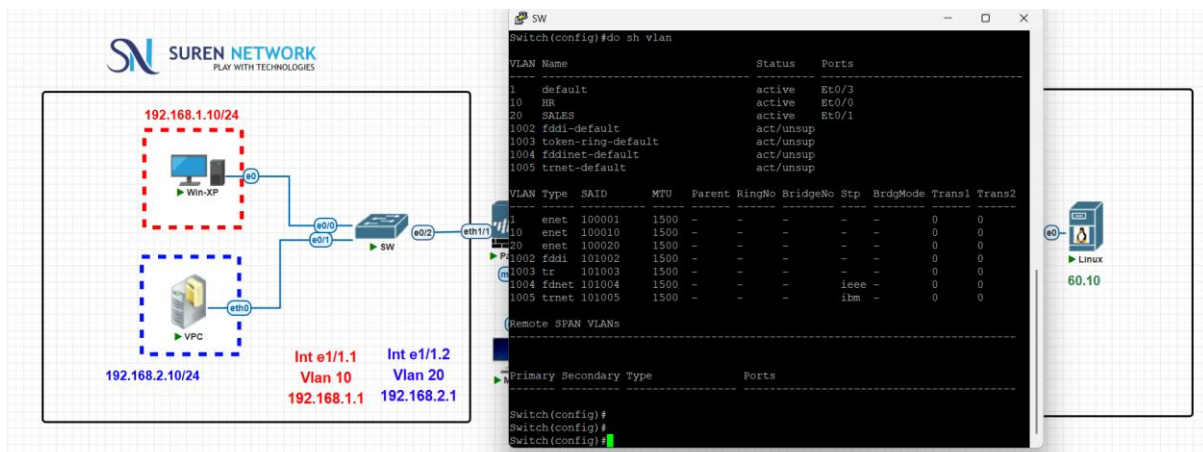
TASK 2: MANAGEMENT PC IP ASSIGNMENT

MGMT PC IP: 1.0.0.10

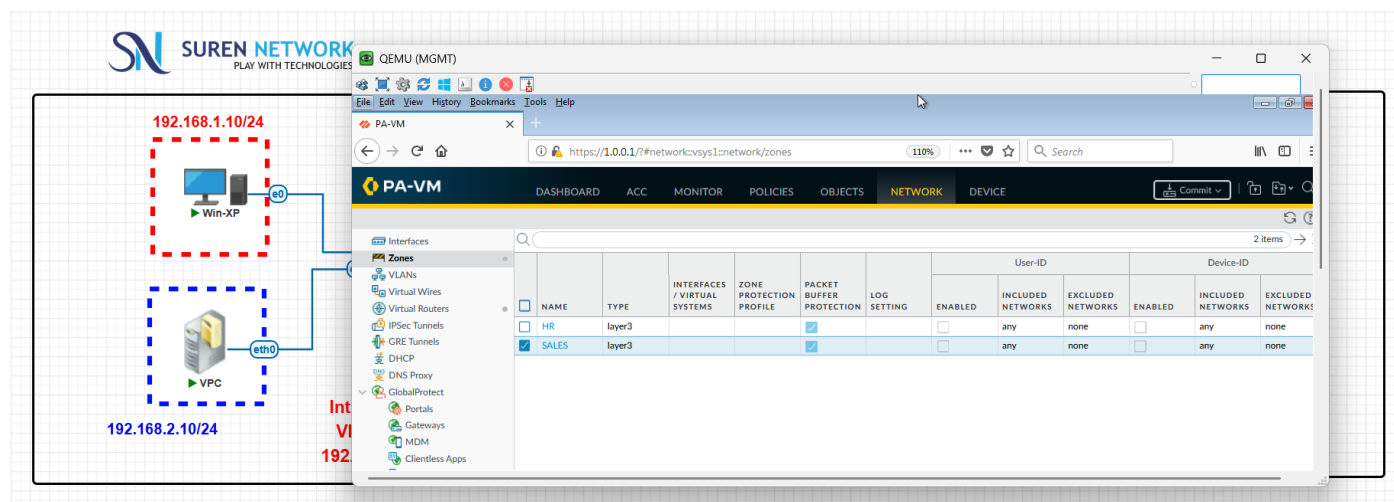
SUBNET MASK: 255.0.0.0



TASK 3: VLAN CONFIGURATION IN SWITCH



TASK 4: CONFIGURING LAN ZONES IN THE NETWORK



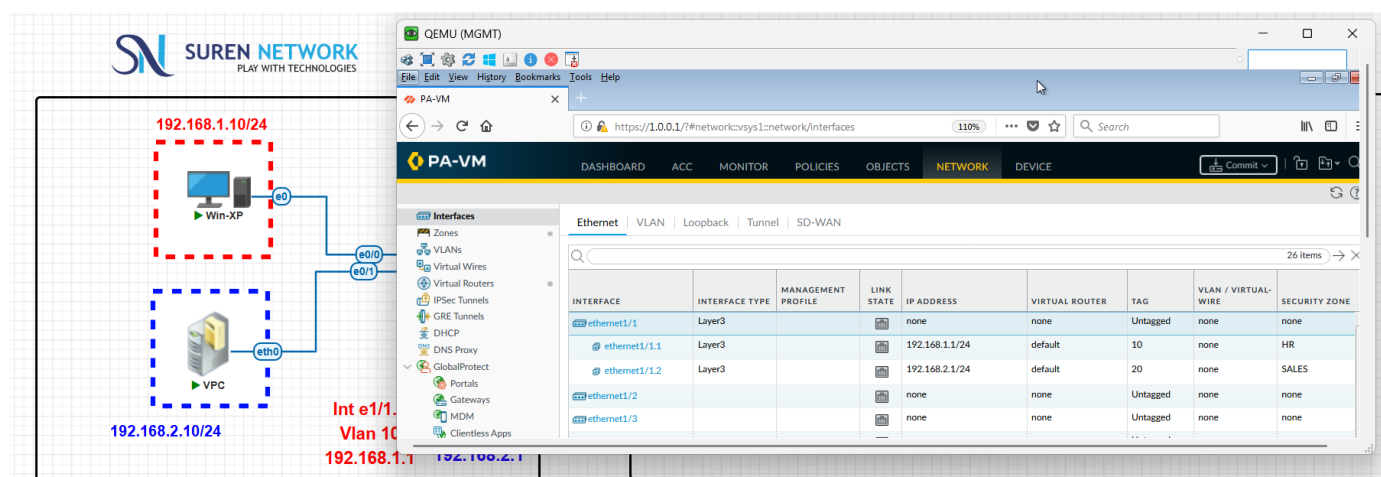
Network Diagram:

- Win-XP (192.168.1.10/24) connected to e0
- VPC (192.168.2.10/24) connected to eth0
- Central switch (Int VI 192.168.1.1)

QEMU (MGMT) - PA-VM - ZONES

NAME	TYPE	INTERFACES / VIRTUAL SYSTEMS	ZONE PROTECTION PROFILE	PACKET BUFFER PROTECTION	LOG SETTING	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS
HR	layer3			☒		☐	any	none	☐	any	none
SALES	layer3			☒		☐	any	none	☐	any	none

TASK 5: SETTING UP INTERFACES (IP Assignments, Default Virtual Routers)



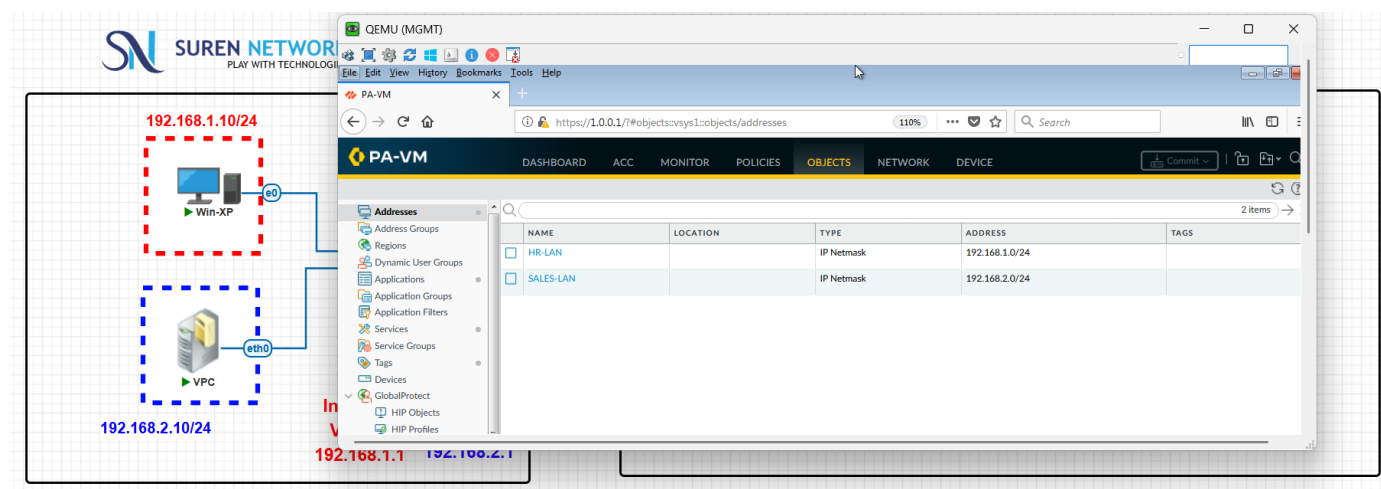
Network Diagram:

- Win-XP (192.168.1.10/24) connected to e0
- VPC (192.168.2.10/24) connected to eth0
- Central switch (Int e1/1, Vlan 10, 192.168.1.1, 192.168.2.1)

QEMU (MGMT) - PA-VM - INTERFACES

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE
ethernet1/1	Layer3		☐	none	none	Untagged	none	HR
ethernet1/1.1	Layer3		☐	192.168.1.1/24	default	10	none	SALES
ethernet1/1.2	Layer3		☐	192.168.2.1/24	default	20	none	none
ethernet1/2			☐	none	none	Untagged	none	none
ethernet1/3			☐	none	none	Untagged	none	none

TASK 6: CREATING NETWORK OBJECT (HR-LAN, SALES-LAN) FOR LAN NETWORK



Network Diagram:

- Win-XP (192.168.1.10/24) connected to e0
- VPC (192.168.2.10/24) connected to eth0
- Central switch (Int e1/1, Vlan 10, 192.168.1.1, 192.168.2.1)

QEMU (MGMT) - PA-VM - ADDRESSES

NAME	LOCATION	TYPE	ADDRESS	TAGS
HR-LAN		IP Netmask	192.168.1.0/24	
SALES-LAN		IP Netmask	192.168.2.0/24	

TASK 7: DRAFTING SECURITY POLICIES FOR LAN TO REACH INTERNET

The network diagram shows a Win-XP VM connected to a VPC via a bridge. The Win-XP VM has IP 192.168.1.10/24. The VPC has IP 192.168.2.10/24. The bridge is connected to the Internet (eth0/1). The VPC is connected to the Internet via a bridge. The VPC is connected to the Internet via a bridge.

The QEMU (MGMT) interface shows the PA-VM dashboard. The Security tab is selected, displaying a table of security policies:

NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE
1 COMMUNICATION	none	universal	HR	HR-LAN	any	any	HR	HR-LAN	any
2 intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any
3 interzone-default	none	interzone	any	any	any	any	any	any	any

TASK 8: PC IP ASSIGNMENT

PC1 IP: 192.168.1.10

PC2 IP: 192.168.2.10

The first screenshot shows the Windows XP network settings window. The 'Internet Protocol (TCP/IP) Properties' dialog is open, showing the IP address 192.168.1.10 and subnet mask 255.255.255.0. The 'Other TCP/IP Settings' tab is also visible.

The second screenshot shows the VPC terminal output. The terminal displays the VPC configuration and the command to add the IP address 192.168.2.10 to the VPC interface.

TASK 9: CONNECTION ESTABLISHMENT VERIFICATION USING ping

The first screenshot shows the Windows XP command prompt. The user has entered the command 'ping 192.168.2.10' and the output shows successful ping results.

The second screenshot shows the VPC terminal output. The terminal displays the VPC configuration and the command to ping 192.168.1.10. The output shows successful ping results.

TASK 10: CONFIGURING WAN ZONES IN THE NETWORK

NAME	TYPE	INTERFACES / VIRTUAL SYSTEMS	ZONE PROTECTION PROFILE	PACKET BUFFER PROTECTION	LOG SETTING	User-ID	Device-ID
HR	layer3	ethernet1/11		☒		any	any
SALES	layer3	ethernet1/12		☒		any	any
WAN 1	layer3			☒		any	any
WAN 2	layer3			☒		any	any

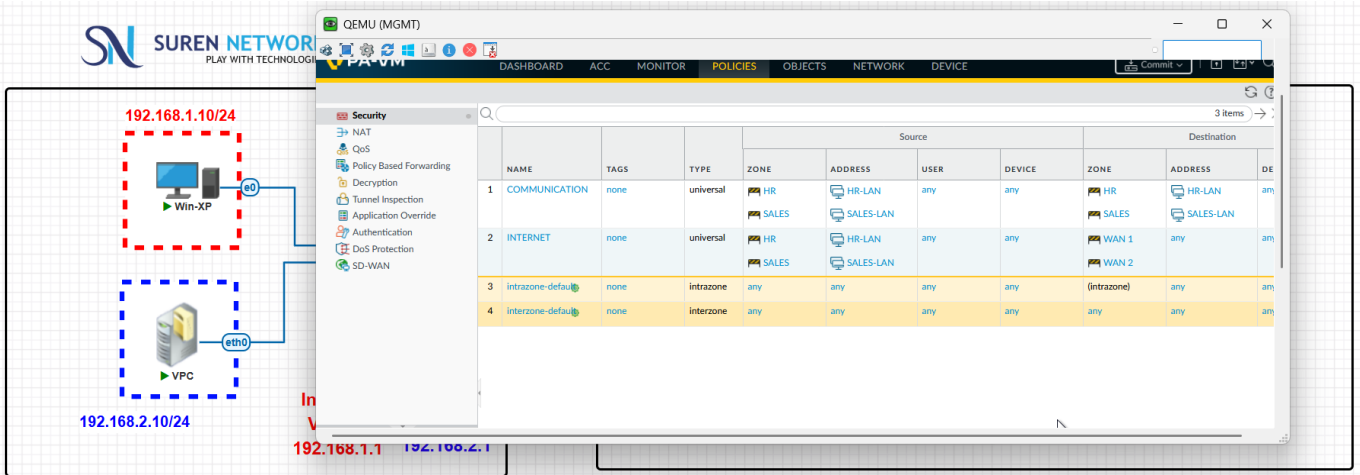
TASK 12: SETTING UP WAN INTERFACES (IP Assignments, Default Virtual Routers)

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE
ethernet1/1	Layer3		☒	none	none	Untagged	none	none
ethernet1/1.1	Layer3		☒	192.168.1.1/24	default	10	none	HR
ethernet1/1.2	Layer3		☒	192.168.2.1/24	default	20	none	SALES
ethernet1/2	Layer3		☒	20.0.0.1/8	default	Untagged	none	WAN 1
ethernet1/3	Layer3		☒	30.0.0.1/8	default	Untagged	none	WAN 2
ethernet1/4	Layer3		☒	none	none	Untagged	none	none

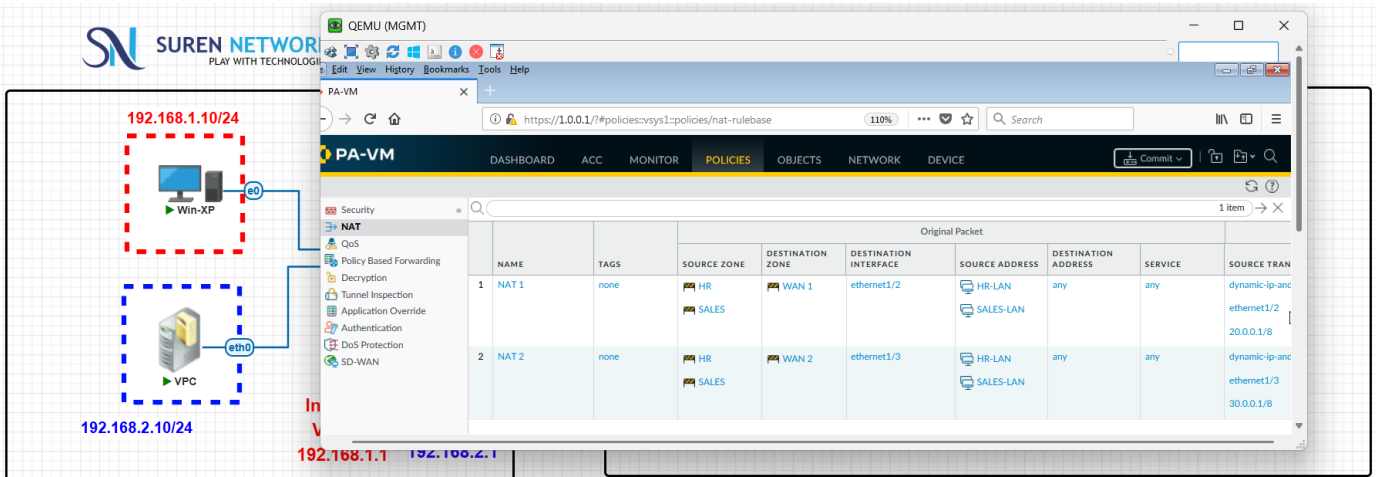
TASK 13: ASSIGNING DEFAULT ROUTES TO ISP 1 AND ISP 2

NAME	DESTINA...	INTERFA...	TYPE	VALUE	ADMIN DISTANCE	METRIC	BFD	ROUTE TABLE
ISP-1	0.0.0.0/0	ethernet1...	ip-address	20.0.0.2	default	10	None	unicast
ISP-2	0.0.0.0/0	ethernet1...	ip-address	30.0.0.2	default	20	None	unicast

TASK 14: DRAFTING SECURITY POLICIES FOR WAN TO REACH INTERNET



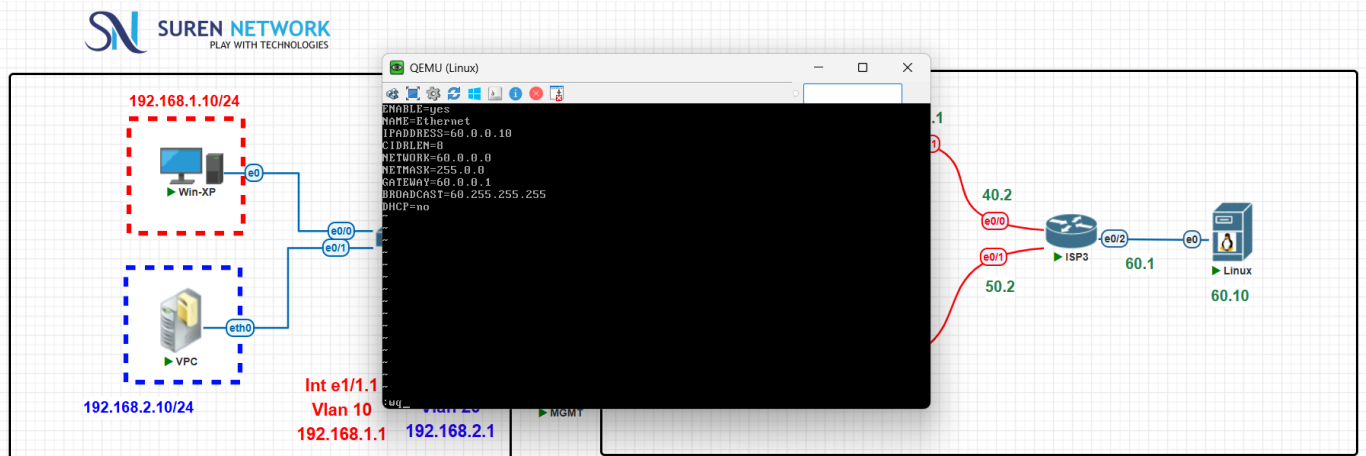
TASK 15: ENABLING NAT POLICIES FOR WAN COMMUNICATION



TASK 16: ASSIGNING IP TO LINUX SERVER

SERVER IP: 60.0.0.10

SUBNET MASK: 255.0.0.0



The screenshot shows a QEMU virtual machine running Windows XP. The command prompt displays the following output:

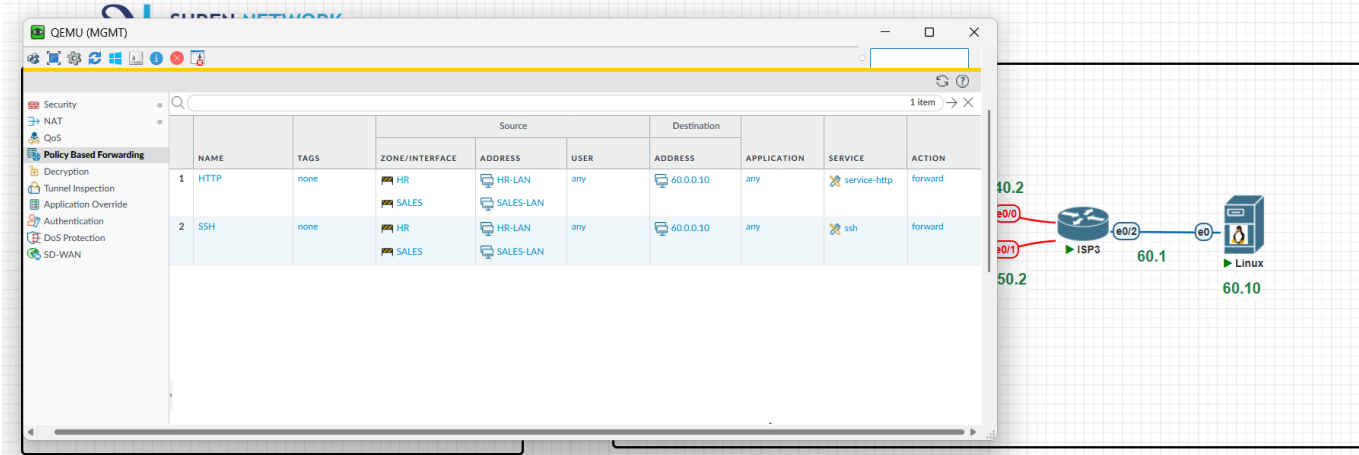
```

C:\WINDOWS\system32\cmd.exe
Reply from 60.0.0.10: bytes=32 time=3ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=3ms TTL=61
Reply from 60.0.0.10: bytes=32 time=5ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=3ms TTL=61
Reply from 60.0.0.10: bytes=32 time=3ms TTL=61
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 60.0.0.10: bytes=32 time=6ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=3ms TTL=61
Reply from 60.0.0.10: bytes=32 time=3ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=4ms TTL=61
Reply from 60.0.0.10: bytes=32 time=5ms TTL=61

```

To the right of the command prompt, a network diagram is shown. It features a central switch labeled "ISP3" with a green triangle icon. The switch is connected to three ISPs (ISP1, ISP2, and ISP3) and a Linux server. The ISPs are labeled with IP addresses 40.1, 40.2, 50.1, 50.2, and 60.1. The Linux server is labeled 60.10. The diagram also shows a "TATA" logo and a "FIBER NET" logo.

TASK 20: ADDING POLICY BASED FORWARDING POLICIES

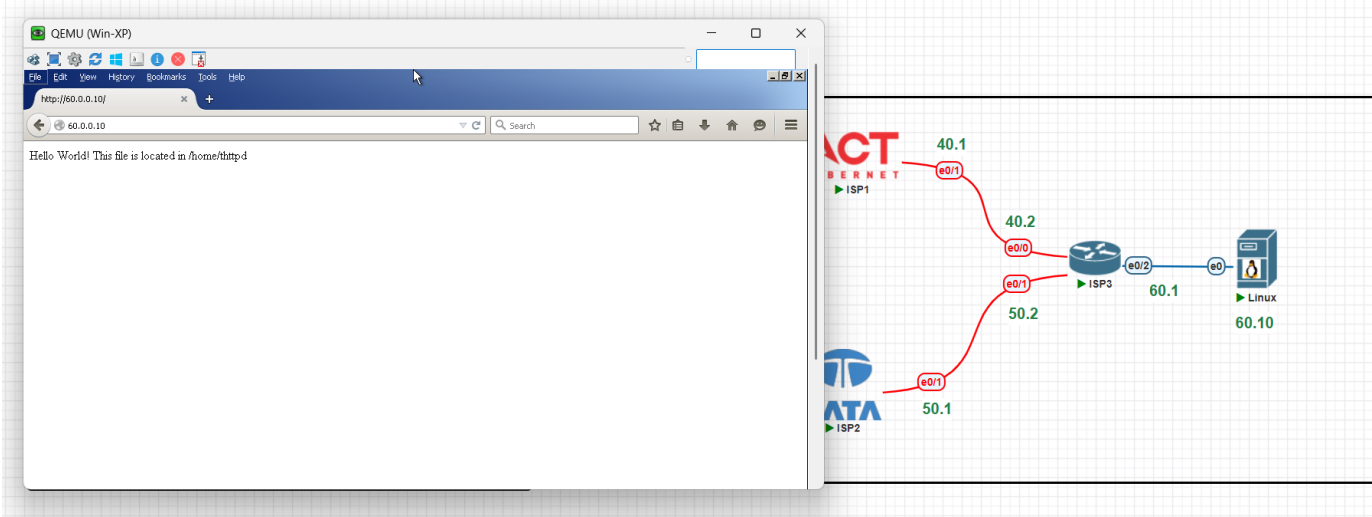


The screenshot shows the QEMU (MGMT) interface with the Policy Based Forwarding configuration window open. The configuration table is as follows:

	NAME	TAGS	ZONE/INTERFACE	ADDRESS	USER	ADDRESS	APPLICATION	SERVICE	ACTION
1	HTTP	none	HR	HR-LAN	any	60.0.0.10	any	service-http	forward
2	SSH	none	HR	HR-LAN	any	60.0.0.10	any	ssh	forward

To the right, a network diagram shows a router (ISP3) connected to a Linux host (60.10). The router has interfaces e0/0, e0/1, and e0/2. The Linux host has interface e0. The router's e0/0 is connected to 40.2, e0/1 to 50.2, and e0/2 to 60.1.

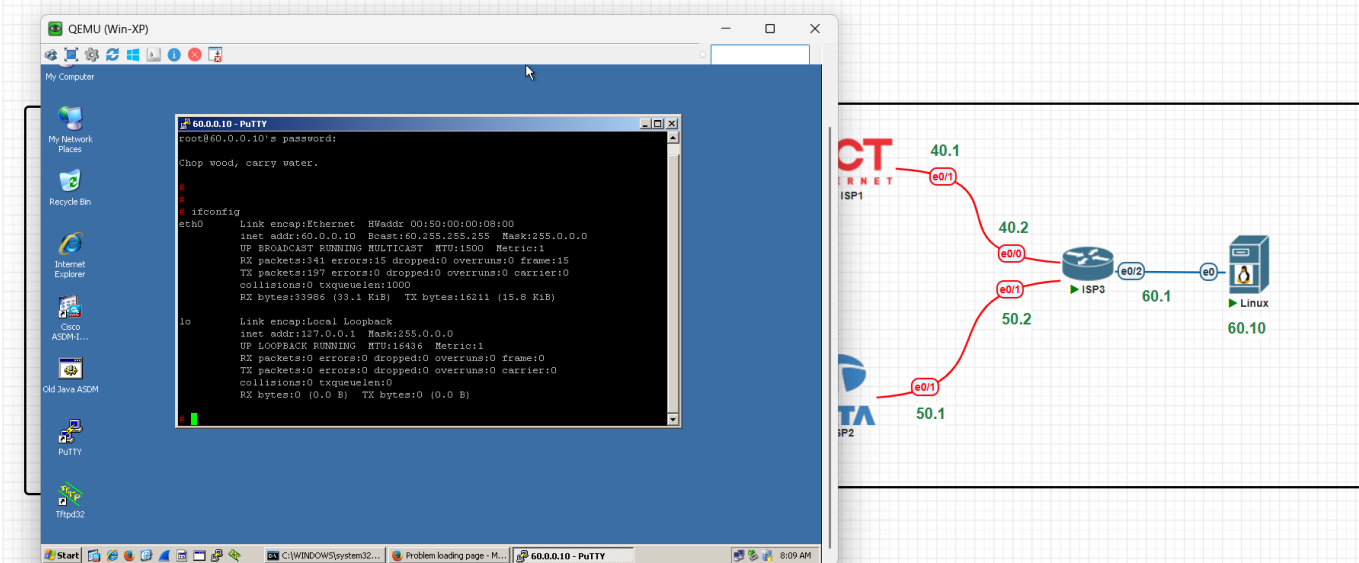
TASK 21: VERIFYING HTTP FROM WEB BROWSER USING POLICY BASED FORWARDING POLICIES



The screenshot shows the QEMU (Win-XP) interface with a web browser open. The browser address bar shows "http://60.0.0.10/" and the page content displays "Hello World! This file is located in /home/httpd".

To the right, a network diagram shows a router (ISP3) connected to a Linux host (60.10). The router has interfaces e0/0, e0/1, and e0/2. The Linux host has interface e0. The router's e0/0 is connected to 40.2, e0/1 to 50.2, and e0/2 to 60.1. The diagram also shows two external ISPs: ISP1 (40.1) connected to 40.2 and ISP2 (50.1) connected to 50.2.

TASK 22: VERIFYING SSH FROM PUTTY USING POLICY BASED FORWARDING POLICIES



The screenshot shows the QEMU (Win-XP) interface with a Putty terminal window open. The terminal displays the following output:

```
root@60.0.0.10's password:
Chop wood, carry water.
#
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:08:00
          inet addr:60.0.0.10  Bcast:60.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:341 errors:15 dropped:0 overruns:0 frame:15
          TX packets:197 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:33986 (33.1 KiB)  TX bytes:16211 (15.8 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

To the right, a network diagram shows a router (ISP3) connected to a Linux host (60.10). The router has interfaces e0/0, e0/1, and e0/2. The Linux host has interface e0. The router's e0/0 is connected to 40.2, e0/1 to 50.2, and e0/2 to 60.1. The diagram also shows two external ISPs: ISP1 (40.1) connected to 40.2 and ISP2 (50.1) connected to 50.2.